Rami Al-Belmpeisi

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ramialbel.github.io/Webpage • GitHub • Google Scholar

Positions Held

Visiting PhD Student, UiT The Arctic University of Norway, Tromsø, NO 2025 – 2026
PhD Student, DTU Compute, Technical University of Denmark, DK 2023 – 2026

Education

PhD in Applied Mathematics and Computer Science, Technical University of Denmark, DK 2026
MSc in Astrophysics, Niels Bohr Institute, University of Copenhagen, DK 2022
BSc in Physics, Aristotle University of Thessaloniki, GR 2020
Erasmus+ Exchange, Eberhard Karls Universität Tübingen, DE 2018 – 2019

Grants & Awards

Winning team, programming challenge, PhD Summer School on Uncertainty in ML, DTU 2026
Travel grant for a research stay at UiT, Otto Mønsted Foundation, DK 2025
First prize, poster competition, HAMLET-PHYSICS, Niels Bohr Institute, DK 2024
PhD scholarship, Novo Nordisk STAR PhD Programme, DK 2023
BSc in Physics in the top 1% of graduates (9.00/10), Aristotle University of Thessaloniki 2020
"The Great Moment for Education" award, Eurobank, GR 2015
12th place, Greek Annual Physics Contest, GR 2013

Publications

1. **R. Al-Belmpeisi**, K. Wickstrøm, J. V. Sundgaard, P. H. Larsen, A. B. Dahl, R. Jenssen, M. Kampffmeyer. RESHAPE: Representation Learning for the Explainability of Shapes. *iMIMIC, MICCAI 2026 Workshops*, LNCS, Springer, 2026 (in press).
2. **R. Al-Belmpeisi**, J. V. Sundgaard, P. H. Larsen, A. B. Dahl. Deep Learning for Liver Disease Stratification: Findings from UKBB MRI. *MIDL*, PMLR 315:4384–4400, 2026.
3. **R. Al-Belmpeisi**, K. A. Sørensen, J. V. Sundgaard, P. Nabilou, M. J. Emerson, P. H. Larsen, L. L. Gluud, T. L. Andersen, A. B. Dahl. Identifying Nonalcoholic Fatty Liver Disease and Advanced Liver Fibrosis from MRI in UK Biobank. *MLMI, MICCAI 2024 Workshops*, LNCS 15242:222–231, Springer, 2024 (oral).
4. **R. Al-Belmpeisi**, V. Tuhtan, M. B. Christensen, R. Kuruwita, T. Haugbølle. Simulated analogues II: a new methodology for non-parametric matching of models to observations. *MNRAS* 534(4):3194–3210, 2024.
5. V. Tuhtan, **R. Al-Belmpeisi**, M. B. Christensen, R. Kuruwita, T. Haugbølle. Simulated analogues I: apparent and physical evolution of young binary protostellar systems. *MNRAS* 534(4):3176–3193, 2024.

Preprint

6. A. I. Yfantis, **R. Al-Belmpeisi**. A deep learning algorithm for black hole spin estimation using hot-spot secondary images. *arXiv:2608.18208*, 2026.

Teaching & Computing

Super-TA, 02002/02003 Computer Programming (Python), DTU Compute, DK
Python (PyTorch, distributed training), C++, CUDA, MPI; RAMSES, PLUTO, RADMC-3D